

Rilzabrutinib Reduces Hives Severity and Frequency in Participants With Chronic Spontaneous Urticaria in RILECSU Phase 2 Trial

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Key Conclusions

- 1** Significant and sustained improvements in UAS7, measuring hives and itch, were observed through Week 12 with rilzabrutinib 1200 mg/day
- 2** Nominally significant improvements in percent change in HSS7 were observed as early as Week 1 with all rilzabrutinib doses. Improvements were sustained through Week 12 with rilzabrutinib 1200 mg/day
- 3** Participants treated with rilzabrutinib 1200 mg/day had more days without hives during the 12-week double-blind period than those receiving placebo
- 4** Rilzabrutinib may be an effective treatment option to reduce the severity and frequency of hives for patients with CSU

Introduction

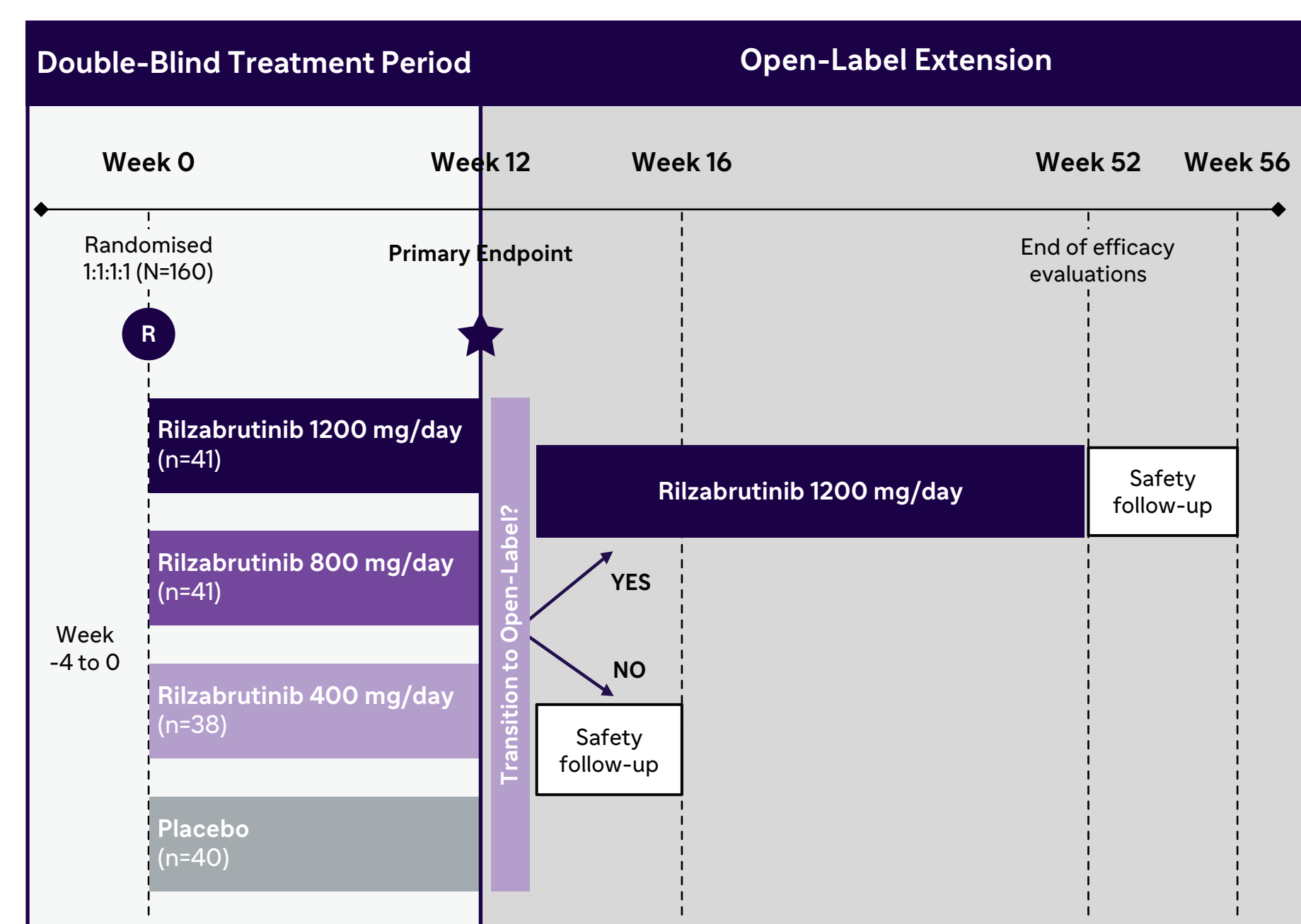
- Chronic spontaneous urticaria (CSU) is a common, chronic, immunological skin disease characterised by the spontaneous appearance of itchy hives, angioedema, or both for >6 weeks¹
- CSU is driven by the pathologic activation of cutaneous mast cells by various mechanisms, such as auto-antibodies²
- Bruton's tyrosine kinase (BTK) is expressed in many immune cells (eg, B cells and mast cells) and plays a critical role in multiple immune-mediated diseases³
- Rilzabrutinib is an oral, reversible, covalent BTK inhibitor⁴
- In the RILECSU phase 2 trial, in patients with CSU whose disease is not adequately controlled with H1-antihistamines, rilzabrutinib 1200 mg/day significantly reduced weekly Urticaria Activity Score (UAS7) vs placebo at Week 12, meeting its primary endpoint⁵
 - Weekly Hives Severity Score (HSS7) was nominally significantly reduced at Week 12 with rilzabrutinib 1200 mg/day

Objective

- To evaluate the effect of rilzabrutinib on the severity and frequency of hives during the 12-week double-blind period of the RILECSU phase 2 trial in adults with moderate-to-severe CSU who were omalizumab-naïve (primary analysis population)

Methods

Figure 1. RILECSU Phase 2 trial (NCT05107115)



Primary analysis population: omalizumab-naïve patients (N=143)
Intent-to-treat population: omalizumab-naïve and omalizumab-incomplete responders (N=160); omalizumab-incomplete responders randomised as follows: 400 mg/day arm (n=1), 800 mg/day (n=6), 1200 mg/day (n=6), and placebo (n=4).
 Note: Rilzabrutinib tablets were taken orally.

Outcomes

- CFB in weekly Hives Severity Score (HSS7) at Week 12
- CFB in HSS7 over time
- Days without hives during 12-week double-blind period (post hoc)

Statistical Analysis

- Multiplicity adjustments were only performed on the primary endpoint

References

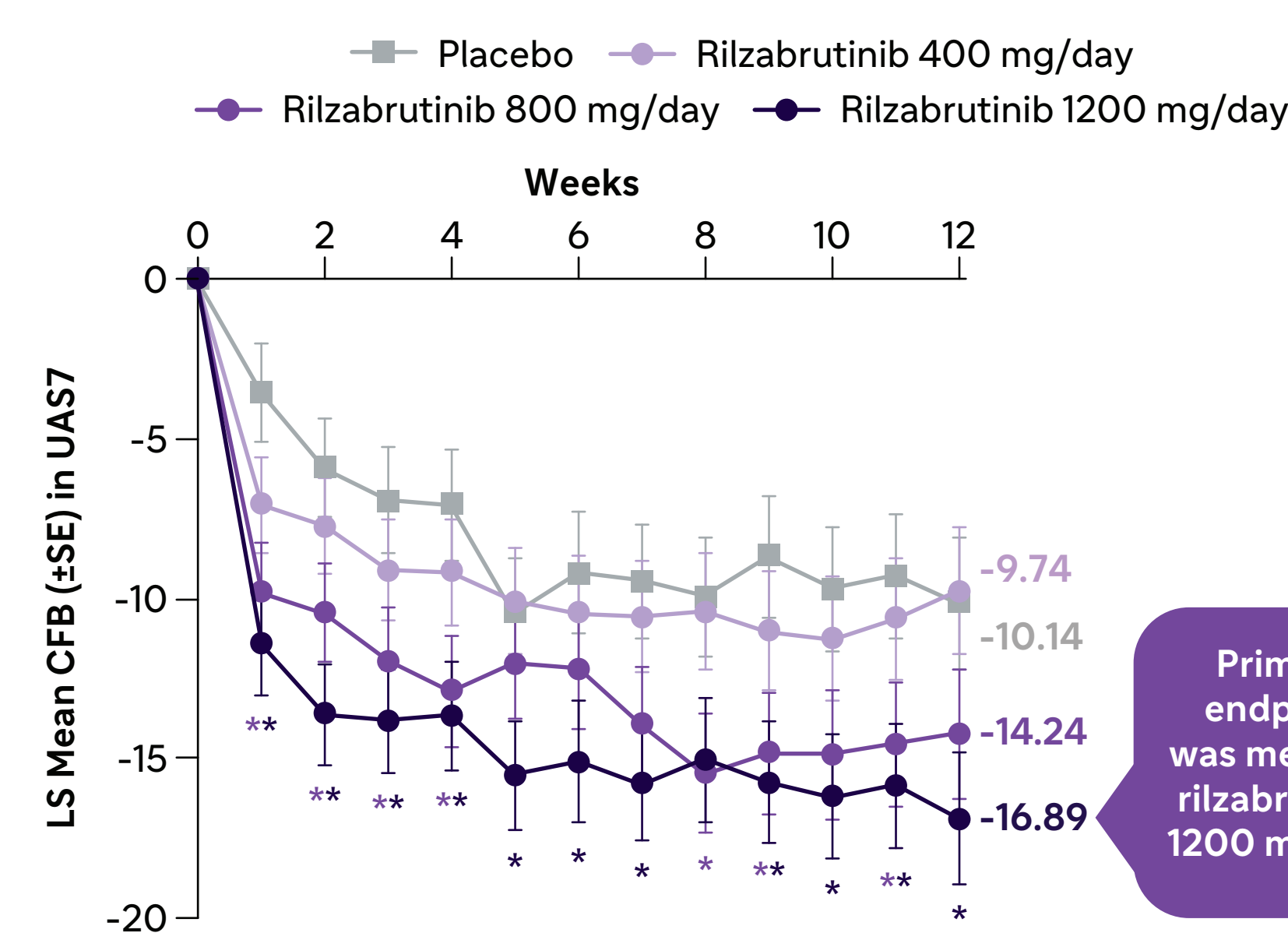
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Disclosures

Silvia Ferrucci is the principal investigator in a clinical trial of Amgen, Sanofi, Novartis, Eli Lilly, LEO Pharma, and AbbVie, and she is an advisory board speaker for Novartis, Menarini, Sanofi, AbbVie, and LEO Pharma. Jordan Talia reports serving as a consultant and/or advisory board member for Arcutis Biotherapeutics, Stifel Financial, LEO Pharma, Johnson & Johnson, Primus Pharmaceuticals, Calliditas Pharmaceuticals, UCB. Gary Gross is a principal investigator in clinical trials for AstraZeneca, GSK, Genentech, Insmed, and Sanofi; has received lecture fees from ALK, AstraZeneca, Sanofi/Regeneron; and is a consultant and advisory board member for Hikma. Akiko Yagami reports lecture fees from Eli Lilly, Novartis, Sanofi, and Taiho Pharmaceutical; and has been consigned/joint research expenses by Maruho. Iris Sun, Leda Mannent, and Jessica Gereige are employees of Sanofi and may hold stock and/or stock options in the company.

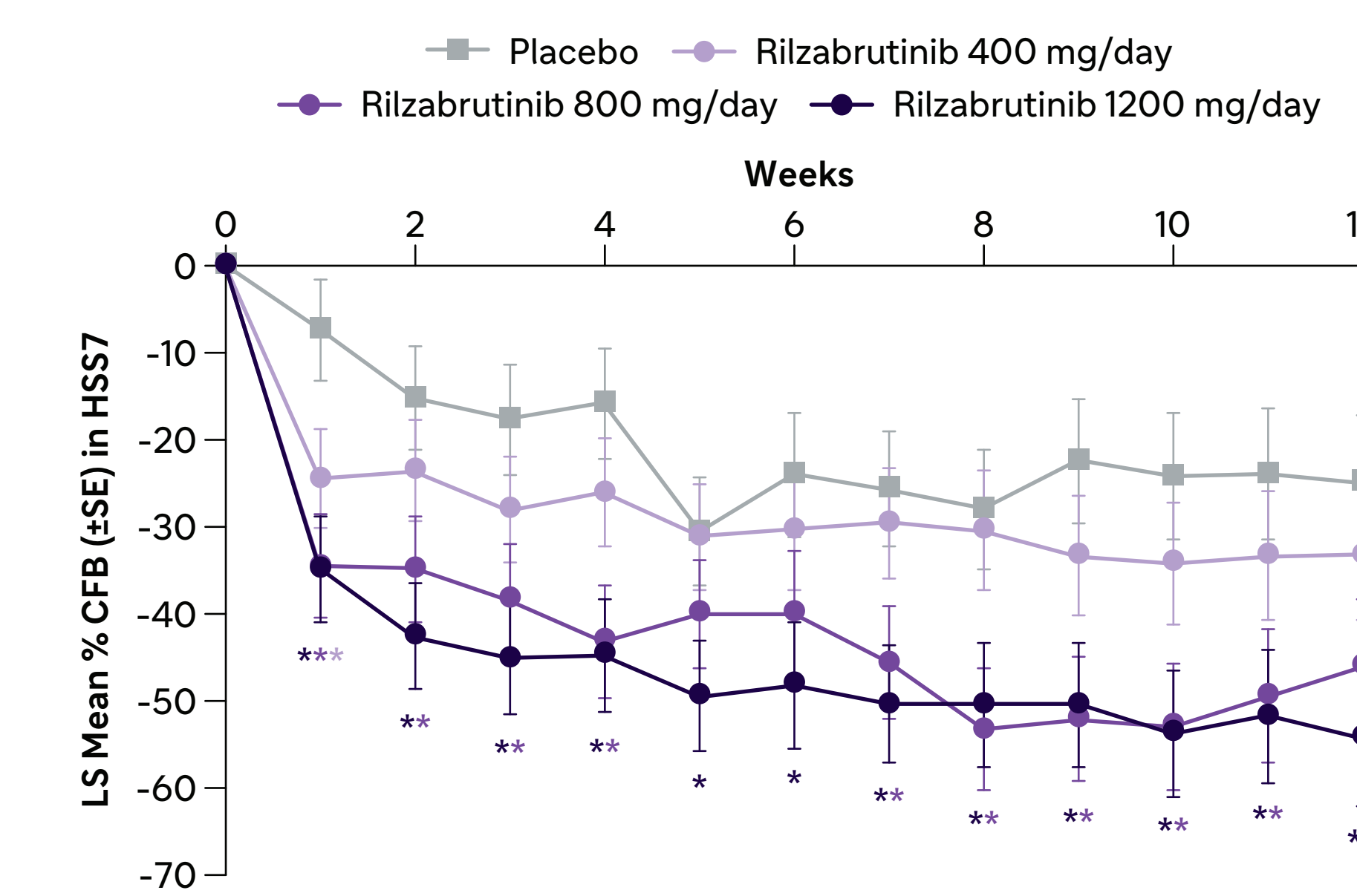
Results

Figure 2. Rilzabrutinib 1200 mg/day showed fast (Week 1) and sustained improvements vs placebo up to Week 12 in UAS7^{6,a}



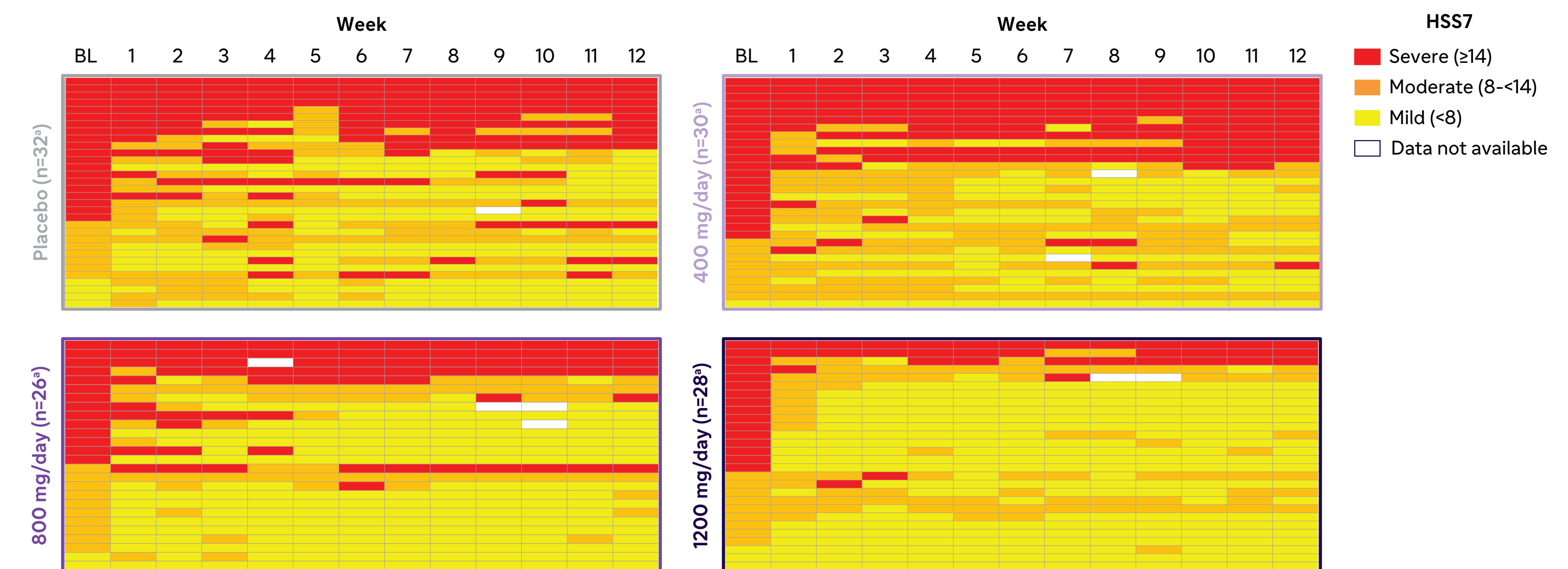
*P<0.05 vs placebo; color-coded based on dose.
 *P-values are nominal except at Week 12.

Figure 3. Rilzabrutinib 1200 mg/day showed fast (Week 1) and sustained improvements vs placebo up to Week 12 in HSS7^a



*P<0.05 vs placebo; color-coded based on dose.
 *P-values are nominal.

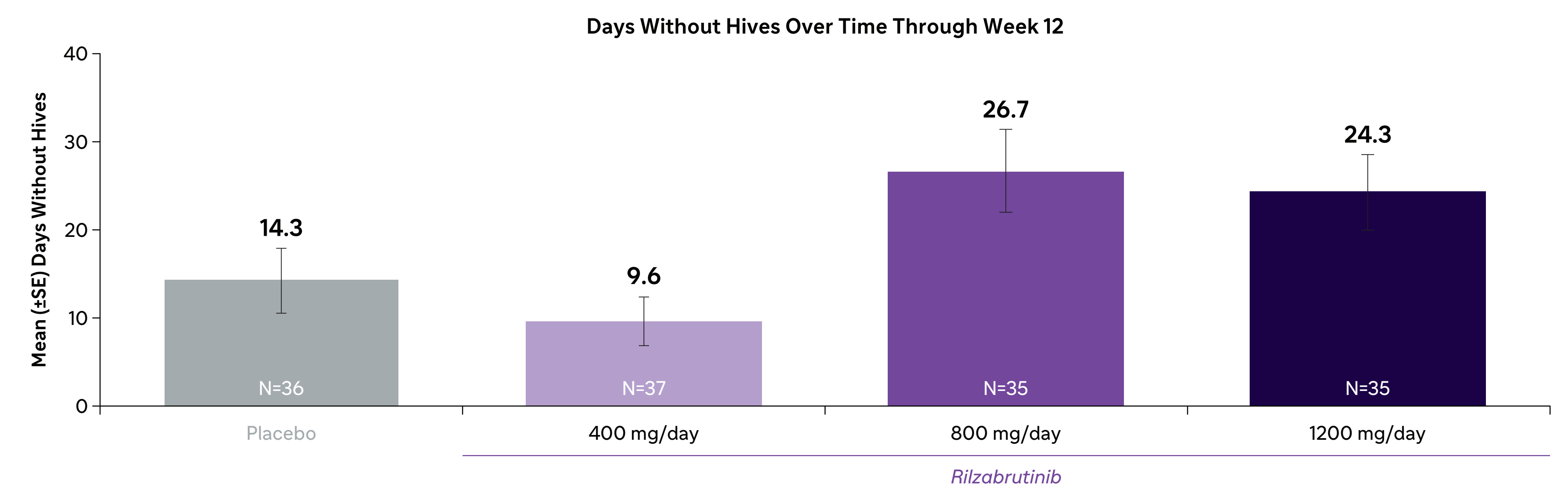
Figure 4. HSS7 severity was improved with rilzabrutinib at Week 12 at an individual level



- No scores worsened with rilzabrutinib 1200 mg/day compared to baseline, and 1 score (3.6%) was unchanged
- Among 16 patients with severe HSS7 at baseline, 13 (81.3%) achieved mild or moderate HSS7 at Week 12 with rilzabrutinib 1200 mg/day

*N is based on primary analysis population (omalizumab-naïve; N=143) with HSS7 scores available at Week 12.

Figure 5. Participants on rilzabrutinib 1200 mg/day experience more days without hives over the 12-week double-blind period than placebo



Abbreviations

BTK, Bruton's tyrosine kinase; CFB, change from baseline; CSU, chronic spontaneous urticaria; HSS7, weekly Hives Severity Score; LS, least squares; MCID, minimal clinically important difference; SE, standard error; UAS7, weekly Urticaria Activity Score.

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